

Dear Students,

Please fill out the form at the link below by **09/09/26**. This form collects information about your tentative app project. Every student must submit the form individually, as each of you is required to develop your own app.

 [Fill out the form here](#)

Guidelines for topic selection:

- Each student must develop an individual app project.
- Topics will be assigned on a first-come, first-served basis — whoever submits the form first gets priority if a topic is proposed by multiple students.
- There are no restrictions on app topics; choose whatever interests you. However, if there's a repeat topic, the later submission must revise theirs.
- You're free to use any platform or programming language.

Required Features

Below is a list of preliminary features your app project must cover. These should be achievable with consistent effort. Failure to implement them will negatively affect your lab grade for this course. Note: except for Crystal Report, all these features were successfully implemented by the previous batch, so feel free to reach out to them for guidance. A few additional features for this semester will be announced soon.

1. Submit your ERD, drawn using Visio, EDrawMax, LaTeX, or any tool that exports a scalable PDF.
2. Real-time commenting that communicates with your own database without refreshing or reloading the page (similar to Facebook).
3. Like/dislike buttons with a live count feature (similar to Facebook).
4. Real-time unique username verification against your own database without page reload.
5. Email and phone number verification, similar to the Gmail registration flow.
6. Manipulation of multiple text boxes and multiple dropdown lists.
7. Interactive real-time rating system for individual posts or for your app/site overall.
8. Pagination.
9. Embedded audio/video playback sourced from an external database (e.g., embedding and playing a YouTube video).
10. Google Maps integration with real-time location data, or at minimum an embedded map iframe.
11. Use of session information (or JWT-based auth) during login/logout.
12. Application of TensorFlow Lite (or an equivalent on-device ML solution) for real-time prediction.
13. Image/file upload functionality with display on your site/app, including cloud storage integration (e.g., Firebase Storage, AWS S3, or Cloudinary).
14. Datepicker.
15. Animations embedded using GSAP or Framer Motion.
16. Application of SASS or Tailwind CSS.
17. Use of modern workflow/collaboration tools such as Git, Gulp.js, Webpack/Vite, or SVN.
18. Application of GraphQL or a modern API layer (e.g., tRPC).
19. RESTful APIs / SOAP APIs.
20. Crystal Report.
21. Use of a modern frontend framework (e.g., React, Next.js, Vue, or Flutter for cross-platform mobile).
22. Basic containerization or deployment pipeline (e.g., Docker, or a CI/CD workflow via GitHub Actions).
23. Integration of a generative AI feature (e.g., a chatbot or content suggestion powered by an LLM API).
24. TBD: additional task-specific features (to be announced).

Please let me know if you have any questions or issues.